



PERGAMON

Annals of Nuclear Energy 27 (2000) 1643–1667

annals of
NUCLEAR ENERGY

www.elsevier.com/locate/anucene

Inelastic neutron scattering and prompt fission neutron spectra for ^{237}Np [☆]

N.V. Kornilov *, A.B. Kagalenko, V.Ya. Baryba,
V.G. Demenkov, S.V. Pupko, P.A. Androsenko

Institute of Physics and Power Engineering, Bondarenko Square 1, 249020 Obninsk, Russia

Received 10 January 2000; accepted 17 February 2000

Abstract

The secondary neutron spectra (inelastic, elastic, fission) for ^{237}Np were measured by the neutron time of flight spectrometer of the IPPE at the incident energy range 1–2.5 MeV. The solid tritium target was used as a neutron source. The neptunium oxide (189 g) packed in the low mass stainless steel container was used as a scattering sample. The neutron background due to scattering on the target environment and tritium into the target backing was measured and was calculated with the appropriate model of the neutron source. The data were corrected for neutron background, the scattering on the oxygen and iron nuclei, and the effect of the finite sample size. The fission neutron spectra were measured, evaluated and subtracted from the emission neutron spectra to estimate inelastic neutron spectra and cross-sections. The experimental results were compared with ENDF/B-VI, BROND-2, JENDL-3 neutron data libraries. © 2000 Elsevier Science Ltd. All rights reserved.

1. Introduction

The ^{237}Np is a dominant nucleus among the nuclei from atomic power waste and the knowledge of the neutron data for this nucleus is of paramount importance for the transmutation problem. Until now there have been no experimental data for inelastic cross-section and scattering neutron spectra that complicates the data evaluation. The first experimental results are presented in this paper.

The measurement of the inelastic neutron spectra and cross-section for ^{237}Np is complicated by the following factors:

[☆] This work was supported by ISTC foundation, project 304.

* Corresponding author. Fax: + 7-095-2302326.

E-mail address: kornilov@ippe.rssi.ru (N.V. Kornilov).

1. A rather massive sample is required for such a measurement, that is, the measurement may be carried out if a 100 g sample of Np is available;
2. The neptunium oxide powder is the only material available for sample preparation. Hence, the detail simulation of the experiment and careful investigation of the neutron source spectra are required for correct estimation of the inelastic neutron spectra and cross-sections;
3. An Np sample produces a very intensive γ -ray background, that should be properly reduced for successful measurements;
4. The total secondary neutron spectra (inelastic, elastic and fission) may be measured for Np. Hence, one should know the spectrum of prompt fission neutrons (PFNS) to estimate the inelastic neutron spectrum. This is a typical problem for fissile nuclei, however, due to the lack of experimental data for Np, additional measurements are required to make a correct interpretation of the PFNS contribution.

In this paper we describe how the above mentioned problems have been overcome. We present only qualitative results in short tables and figures, having in mind that the total set of experimental data is available from the final ISTC report (Kornilov et al., 1997). The paper is organized as follows. In Section 2 we describe the experimental method, results of measurements and model calculations of the neutron spectra from solid tritium target used as a neutron source at energy range 1–2.5 MeV. In Section 3 we present results of the PFNS measurements and evaluation in the incident energy range 0.5–2.5 MeV. The data processing, results of Monte Carlo simulation and comparison of the inelastic neutron spectra are presented in Section 4. We close with some conclusions in Section 5.

2. Experimental method for neutron spectra measurement

2.1. Neutron spectrometer

The neutron spectra were measured by time-of-flight (TOF) technique with the EG-1 accelerator spectrometer of the IPPE (Birukov et al., 1990; Kornilov et al., 1991). The parameters of the spectrometer: pulse width ~ 2 ns, repetition rate 2 MHz, flight path ~ 2 m, mean current on the target 4–6 μ A. The experimental set up is shown in Fig. 1. The T(p,n) reaction in a solid tritium–titanium target was used as a neutron source at energy range 1–2.5 MeV. Neutrons with energy ~ 0.5 MeV were produced with Li(p,n) reaction. In the main the experimental method was applied for ^{235}U , ^{238}U secondary neutron spectra measurement (Kornilov and Kagalenko, 1995).

2.2. Samples

The Np sample was manufactured from NpO_2 powder packed in a stainless steel container. The container is a cylinder with wall thickness 0.1 mm and bottom thickness 0.3 mm. The container weight is 19.9 g. The dimension of the powder

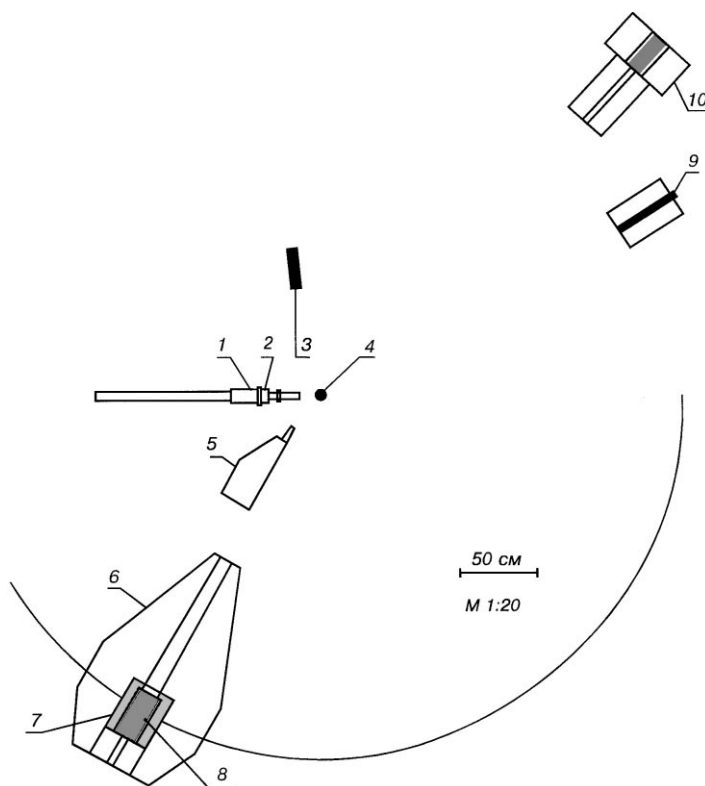


Fig. 1. Experimental setup of an IPPE spectrometer, 1. pick-up electrode; 2. neutron source; 3. TOF monitor for pulse width control; 4. sample; 5. shielding bar; 6. shielding collimator of the main detector; 7. lead shielding; 8. main detector; 9. long counter; 10. TOF monitor for neutron target control.

volume is 45 mm in height and 47 mm in diameter. The NpO_2 weight is 189.1 g. The carbon sample was 47 mm in outer diameter, 27 mm in inner diameter and 45 mm in height. The samples were arranged at a distance 10 cm from the target. The elastic scattering on carbon was used for determination of the neutron flux on the investigated sample.

2.3. Main detector

The neutron detector was a stilbene crystal with 70 mm in diameter and 50 mm thickness. The scintillator was viewed by a photomultiplier tube (PMT) FEU-30. The detector face was covered with lead absorber to 20 mm in thickness to reduce the background of sample γ -rays. In addition, the pulse shape analysis was used to discriminate the γ -rays. The count rate of the sample gamma-rays by the detector placed at ~ 200 cm distance from the sample was $\sim 1.9 \cdot 10^4 \text{ s}^{-1}$ (no any shielding) and $\sim 60 \text{ s}^{-1}$ (20 mm Pb shielding). The detector was shielded with a moveable collimator.

2.4. Monitors

There were three monitors:

1. “long” counter for relative measurements;
2. TOF monitors to check the neutron source parameters and proton pulse width. The monitors consists of plastic scintillator (30 mm in diameter and 10 mm thickness) coupled with PMT FEU-143. One of them (3 in Fig. 1) was placed at the 70 cm distance from the neutron source and another one (10 in Fig. 1) supplying with shielding was placed at ~ 300 cm path flight to measure energy distribution of the neutrons from the target. Due to a high time independent background, this monitor was applied for the estimation of the time resolution and the energy spread of the main component of the neutron source.

2.5. Spectra measurement

All spectra were measured at an angle of 120° with respect to proton beam direction. Each measurement of the Np-sample consisted of 10–12 runs with duration about 1 h. For some initial neutron energies the measurements were repeated. Each sample experiment was accompanied by the measurements of: neutron spectrum with an empty container, neutron spectrum with C-sample and background spectrum (no sample).

For each run the following data arrays were collected: two-dimension distributions (TOF*pulse height (PH)) — 512×128 channels for main detector in coincidence with neutron and gamma-rays (time channel width 1.005 ns), TOF spectrum for monitor (10) — 512 channels (channel width 0.987 ns), TOF spectrum for monitor (3) — 2048 channels (channel width 0.24 ns). In addition the one-dimension spectra from ADC and TDC connected with main detector were collected as well. The two-dimension spectrum for Np-sample is shown in Fig. 2.

2.6. Data processing for raw experimental data

At the first stage of processing the monitor time-of-flight spectra were applied for the verification of the accelerator performance quality (time resolution, energy stability) in each run. The spectra of “good” runs were summed up. The time independent background was subtracted from overall spectra. The detector spectrum was measured as a two-dimension array, therefore the zero-effect time interval was known for each PH channel (upper left part of the spectrum in Fig. 2). Content of channels in these time intervals were reduced to zero after the background subtraction. After this procedure TOF distributions for each PH channels were summed up to calculate the total TOF spectra. This background subtraction resulted in the improvement of measurement accuracy in the low energy range of neutron spectrum. The resulting time spectrum was normalized to the counts of the “long” counter. The TOF spectra for ^{237}Np run and container run after time independent background subtraction are shown in Figs. 3 and 4. Neutron energies increase with increasing of TOF channel number.

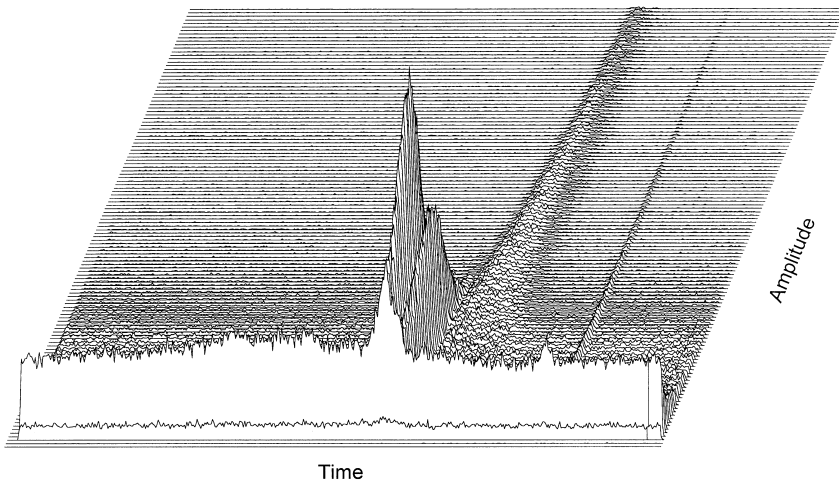


Fig. 2. Two-dimension distribution for the main detector (neutron coincidence mode).

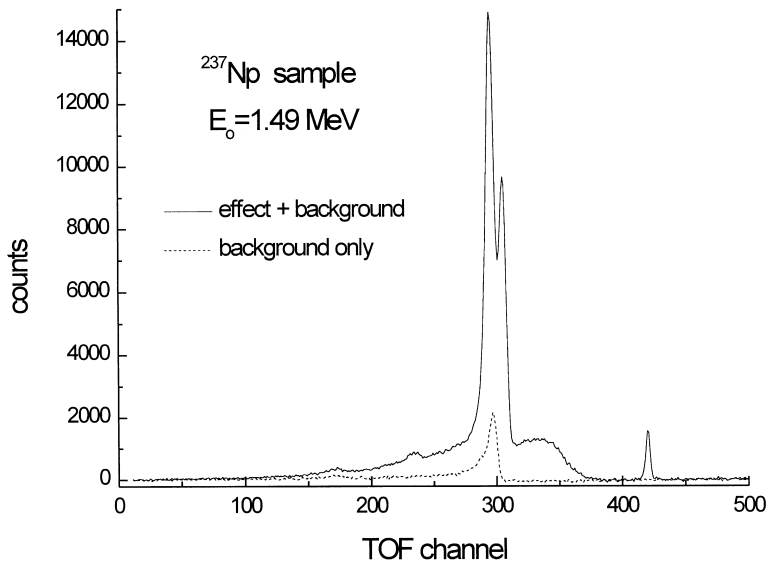


Fig. 3. TOF spectra for “Np sample” run and “no sample” run after the subtraction of time independent background. The spectra are normalized for long counter number.

The peak at ~ 420 channel is due to sample γ -rays (fission and inelastic scattering). The smooth distribution at > 310 channel is produced by fission neutrons. The following peculiarities of the ^{237}Np TOF experimental spectrum should be highlighted.

There are three peaks in the middle of TOF scale. More intensive ones are connected with elastic scattering on Np (higher energies) and O nuclei. The peak at ≈ 240 channel (see also Fig. 3) is due to neutrons inelastically scattered by Fe nuclei.

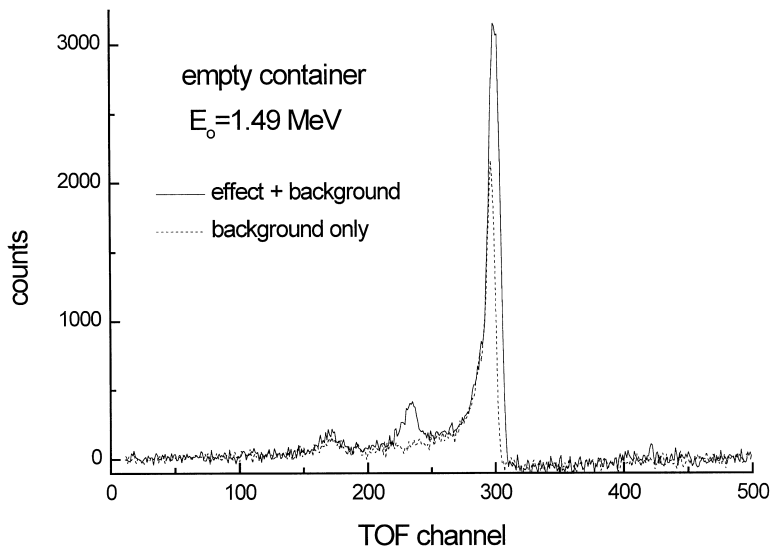


Fig. 4. TOF spectra for “container” run and “no sample” run.

The sample nuclei ratio is 1:2:0.5 (Np:O:Fe), therefore, we cannot use rather simple procedure i.e. subtraction of the container spectrum from sample spectrum run to estimate the net spectrum. Therefore, the contribution of the Fe nuclei (as well O nuclei) should be estimate by Monte Carlo (MC) simulation.

2.7. Investigation of main detector properties

The main detector parameters was measured with the standard ^{252}Cf fission neutron spectrum (Mannhart, 1986), when the fast ionization chamber with ^{252}Cf source (Baryba et al., 1977) was attached in the sample location. Fission fragments count rate was $8.15 \cdot 10^4 \text{ s}^{-1}$. The following parameters were measured in this experiment: detector efficiency, pulse height response and time resolution as a function of neutron energy. The detector efficiency was measured for two high voltage supplies i.e. 1.9 and 1.8 kV (Fig. 5). In the first case the sensitivity to low energy neutrons was increased very much. However, high energy pulses were rejected due to the dynamic range limitation. This mode was applied for inelastic neutron spectra measurements. In latter case we have rather high efficiency for high energy neutrons and this mode was used for measurements of the fission neutron spectra at 0.52 MeV. The detector efficiency was measured several times during the experiment.

The time resolution depends on detector properties (scintillator, PMT, timing discriminator) and pulse height response for separate neutron energy. The time resolution was described by the following expression:

$$\rho(t, t_0) = \sum_{i=1}^{i_{\max}} a_i(E) \rho_i(t, t_0), \quad \sum_{i=1}^{i_{\max}} a_i(E) = 1. \quad (1)$$

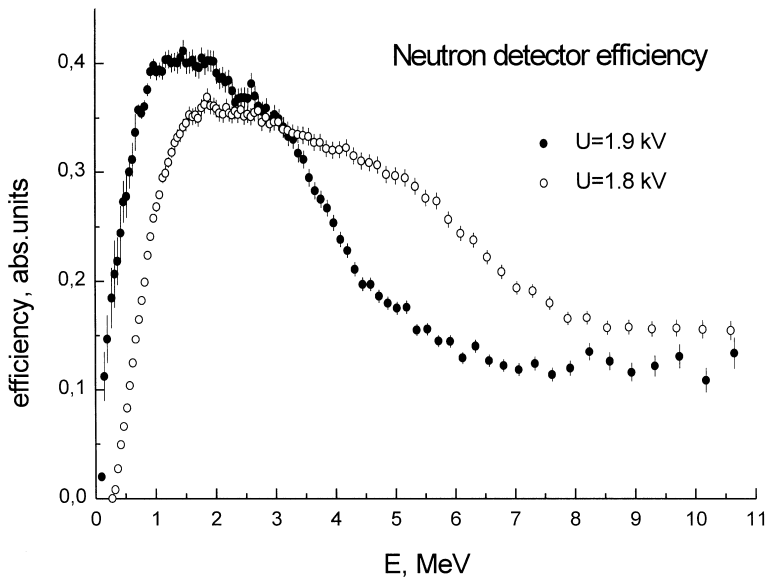


Fig. 5. Main detector efficiency for two high voltage supplies.

where $\rho_i(t, t_0)$ is the time resolution function for i -th pulse height channel, $a_i(E)$ is a pulse height response for neutron energy E , $\rho(t, t_0)$ is the time resolution for neutron energy E . We assumed that $\rho_i(t, t_0)$ and $\rho(t, t_0)$ are the Gauss functions. Hence the second central momentum σ of the $\rho(t, t_0)$ may be calculated with the formula:

$$\sigma^2(E) = \sum_{i=1}^{i_{\max}} a_i(E) \sigma_i^2 \quad (2)$$

Time variance σ_i for each pulse height bin was calculated from γ -peak. So, we assumed also that time resolution for recoil protons is equal to time resolution for electrons if they produce the same pulse height.

Some responses are shown in Fig. 6. The time resolution of the detector versus neutron energy is presented in Fig. 7. The increase at $E > 3$ MeV is connected with the discrimination of high energy amplitudes.

2.8. Neutron source spectrum

The $T(p, n)$ reaction in a solid tritium–titanium target was used as a neutron source at the energy range 1–2.5 MeV. The target thickness was 0.4 mg/cm². The target backing thickness was 0.3 mm. It was cooled by 0.3 mm water layer which was covered by stainless steel can to 0.3 mm. The diameters of these zones were 15 mm.

The energy-angular distributions of the neutrons escaped from the solid tritium target were measured and were calculated by Monte Carlo method. The following factors were taken into account:

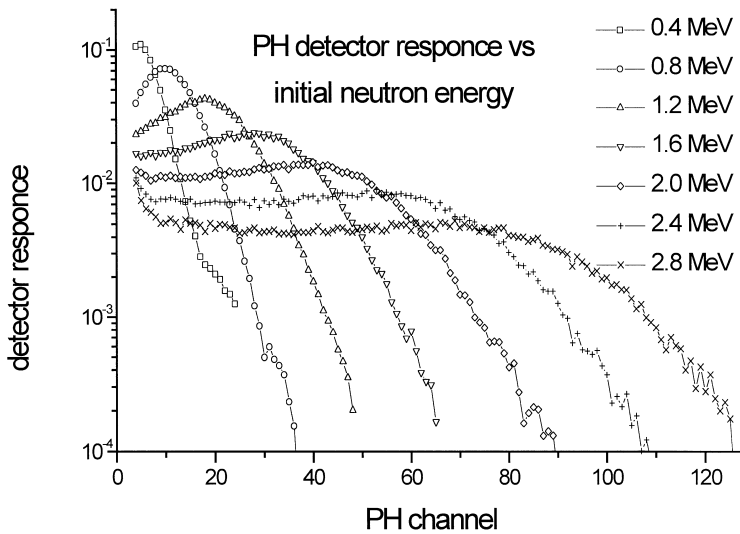


Fig. 6. Detector responses for various neutron energies.

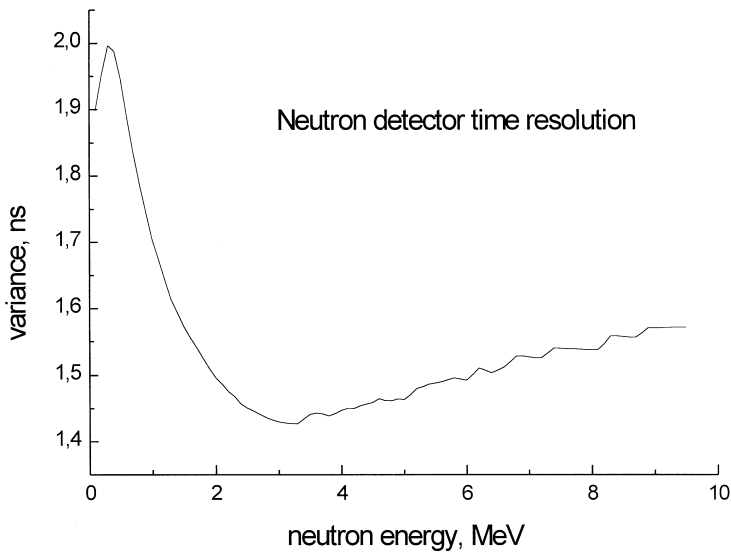


Fig. 7. Time resolution of the detector as function of neutron energy.

- space distribution and energy spread of incident protons;
- energy losses in target (target thickness and its distribution), energy and angle straggling;
- the variations of the neutron energy and reaction cross-section with the angle of the neutron escape;
- neutrons scattering on the target environment;
- neutrons produced by tritium distributed into the target backing.

From the comparison of experimental and MC calculated TOF spectra (Fig. 8) one can conclude that the observed spectrum cannot be described as a sum of “monoenergetic” neutron source (dashed) and neutrons scattered on the target environment (dotted); there is the third (“additional”) neutron source. The similar deviations were demonstrated for various incident energies and target backing (Mo and Cu). It was found, that the intensity and the spectrum shape of these “additional” neutron source can be described if one suppose that there exist some amount of the tritium into the target backing (“backing” tritium). This conclusion is true for both targets. The uniform distribution of this “backing” tritium along the proton range was assumed in our calculation. In spite of this simplification, the calculated results are in quite good agreement with experimental data.

A set of parameters and functions (proton energy losses, neutron energy versus angle) are needed for this calculation. Most of them were taken from well known systematics (see Kornilov, 1986) or were estimated by simple geometric measurements. The other parameters (i.e. the target profile and the part of the “backing” tritium) were taken from the comparison of the main detector and monitor spectra and calculated results.

The neutron scattering on the target environment and “backing” tritium produce the low energy tail accompanying the main neutrons. For our particular target assembly the contribution of the scattered neutrons was $\sim 3.6\%$ at 0° . In the same time, $\sim 20\%$ of the “backing” tritium should be included in our source model to

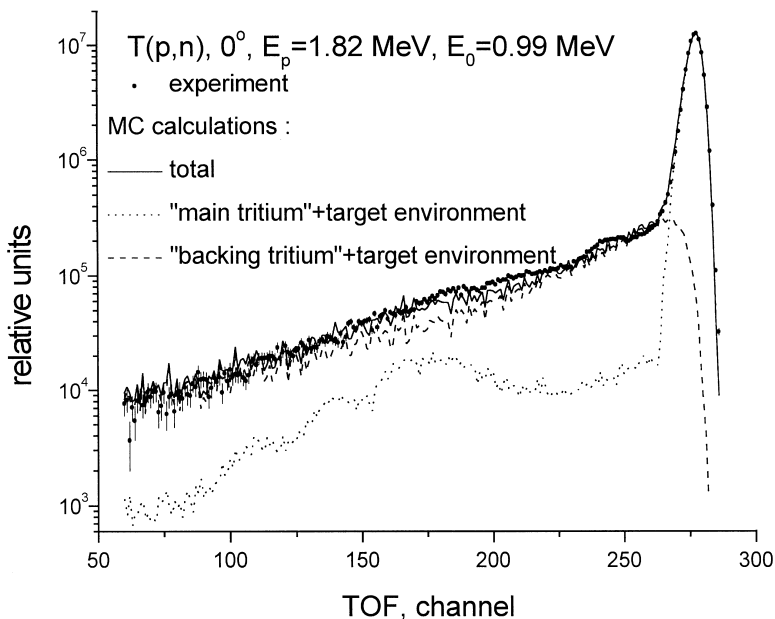


Fig. 8. TOF spectra from the neutron source at 0° angle. Circles are the experimental data, lines show the MC result. The detector efficiency was taken into account in this calculation.

describe the experimental data. This amount of the tritium produces $\sim 11\%$ of the counted neutrons for 1.82 MeV protons.

We demonstrated in this subsection only main detail which are important for data evaluation. An additional information for neutron source spectra, measurement detail and data analysis may be found in Kornilov et al. (1998).

3. Prompt fission neutron spectra

3.1. Spectrum measurement at 0.5 MeV neutron energy

The fission neutron spectrum for ^{237}Np was measured at the incident energy 0.52 MeV. The water-cooled metallic lithium target prepared “in situ” in the accelerator pipe was used as a neutron source (Kornilov and Kagalenko, 1996). The target was evaporated on the stainless steel polished backing 0.5 mm in thick. The target thickness was $\sim 0.6 \text{ mg/cm}^2$. The initial neutron energy 0.52 MeV ($E_p^0 = 2.30 \text{ MeV}$) had been chosen to provide the maximal neutron yield from $\text{Li}(p,n)$ reaction. The following spectra were measured: the neutron spectrum with ^{237}Np -sample and the background spectrum (no sample). Experimental spectra in the time scale are shown in Fig. 9. After the background subtraction experimental spectra were corrected for nuclei admixture and sample size effect. The correction was rather small $\sim (1\text{--}5)\%$ and was as a smooth function in the energy range from 0.7 to 10 MeV.

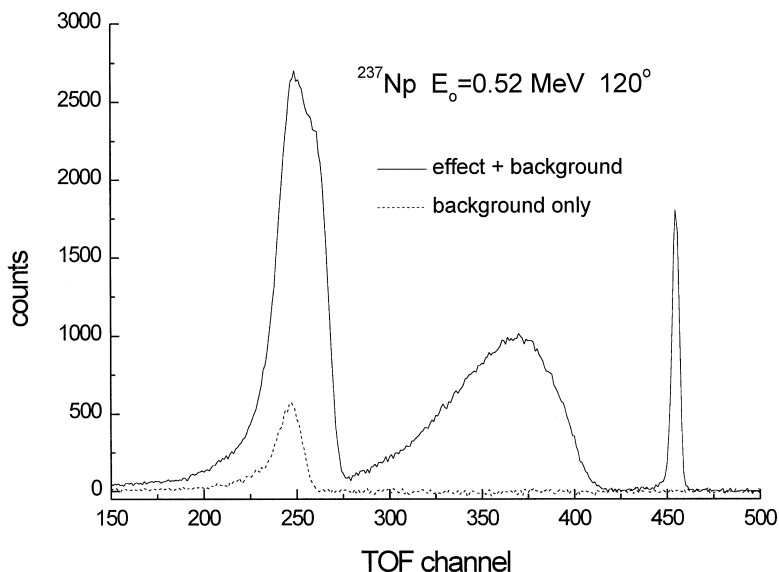


Fig. 9. TOF spectra for “Np sample” run and “no sample” run after the subtraction of the time independent background. The spectra are normalized for “long counter” number.

3.2. Bin-width correction and time resolution correction

The bin-width correction of the experimental data was discussed in Madland and Nix (1982). The authors state that bin-width correction consists in determining a set of energies E_b by which the experimental histogram can be represented with a set of experimental points. They gave the following notations:

$$\int_{E_1}^{E_2} N(E)dE = N_b(E_2 - E_1), N(E_b) = N_b \quad (3)$$

where E_1 and E_2 are lower and upper bin limits, $N(E)$ is true spectrum. Assuming the exponential distribution for $N(E)$ a simple formula was suggested for E_b calculation. However, the same results, that is, the proper correction of the experimental data may be achieved if one fixes the middle point energy E_m and calculates the spectrum magnitude with the equation: $N(E_m) = kN_b$. The “bin-width” correction k may be estimated as a ratio of the simulated time of flight spectrum to the true one. This method is a particular important for spectrum measurement relative ^{252}Cf when the correction (k values are close) may be compensated in the main part. The calculation of the correction including time resolution correction was made by Monte Carlo method. The total correction $\xi(E)$ (the experimental spectrum should be multiplied with this value) is a rather smooth function of energy: $\xi(< 8 \text{ MeV}) = 1$, $\xi(10 \text{ MeV}) = 0.96$, $\xi(13 \text{ MeV}) = 0.91$. The correction for the bin-width is small for our experimental condition $k(13 \text{ MeV}) = 0.996$. In one’s turn the total correction is much less of the experimental errors $dN/N(13 \text{ MeV}) = 0.6$ and was not included in data analysis.

3.3. Analysis of the experimental results

The comparison of experimental results [our for 0.52 MeV and Boikov et al. (1994) for 2.9 MeV] with ENDF/B-VI data (Figs. 10 and 11) shows that ENDF/B-VI spectrum shape can not be applied for PFNS subtraction. For this purpose we used the result of recent systematic and PFNS evaluation. The basement of this approach is presented below referring readers to our paper (Kornilov et al., 1999) for more details, tables of the various parameters and references.

The simplest and most commonly used approximation for the PFN spectra in the laboratory reference system (LS) is Maxwell distribution:

$$M(E, T) = \frac{2\sqrt{E}}{T\sqrt{\pi T}} \exp\left(-\frac{E}{T}\right), \quad (4)$$

If for the shape of the neutron evaporation spectrum in the fragment center of mass system (CMS) a Maxwell distribution (4) is assumed, and furthermore if it is assumed that all fragments have the same kinetic energy per nucleon E_v then the laboratory neutron spectrum shape is so named Watt distribution:

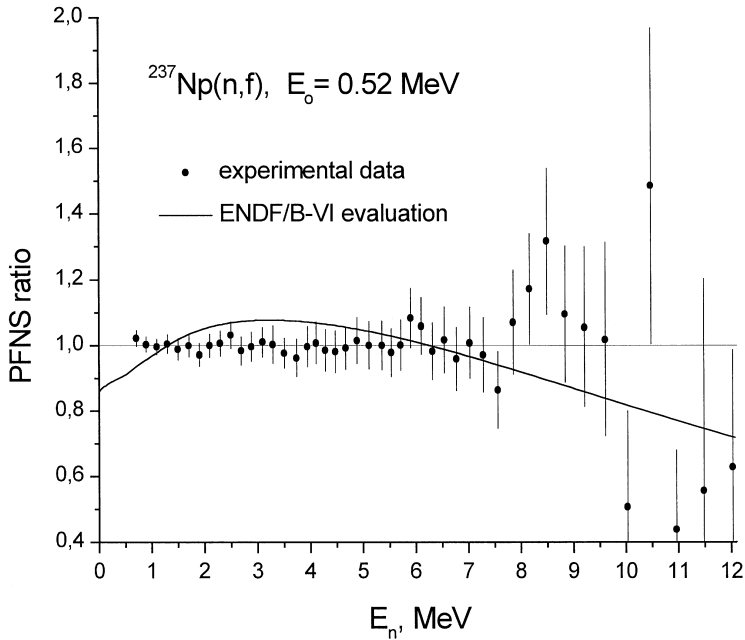


Fig. 10. The ratio of ENDF/B-VI evaluation and results of the present work to our evaluation Kornilov et al., 1999. Residual chi-square per experimental point is $\chi^2/m = 0.22$ and 1.58 (ENDF/B-VI) for energy range 0.7–10 MeV, $m = 40$.

$$W(E, T, E_v) = M(E, T) \exp\left(-\frac{E_v}{T}\right) \frac{\text{sh}(\sqrt{bE})}{\sqrt{bE}}, \quad b = \frac{4E_v}{T^2}, \quad (5)$$

The average neutron energy in the LS is connected with parameters of these equations and the CMS neutrons average energy $\langle \varepsilon \rangle$ by the relations: $\langle E \rangle = 1.5T$ for Eq. (4) and $\langle E \rangle = \langle \varepsilon \rangle + E_v = 1.5T + E_v$ for Eq. (5).

The first approach is a very simple parameterisation. Therefore it cannot describe the PFNS shape in the energy range to ~ 20 MeV with high accuracy. The Watt formula has a more realistic basis. As was obtained by Le Couter and Lang (1959), the CMS neutron spectrum due to the cascade emission process may be described by the formula: $N(E) \sim \varepsilon^\lambda \exp(-\varepsilon/T)$, with $\lambda = 5/11$ for cascade neutron emission. So, it may prove the application of the Maxwellian for CMS spectra calculation. However, the second assumption is wrong, E_v and T depend on mass and kinetic energy of fragments. In addition, Eq. (5) contains two very correlated parameters. As a result the values fitted to experimental spectra have rather large errors that complicate analysis and systematics of the experimental data.

In our evaluation we tried to conserve a very simple analytical approach and avoid the shortcomings mentioned above. Our evaluations are based on the following assumptions.

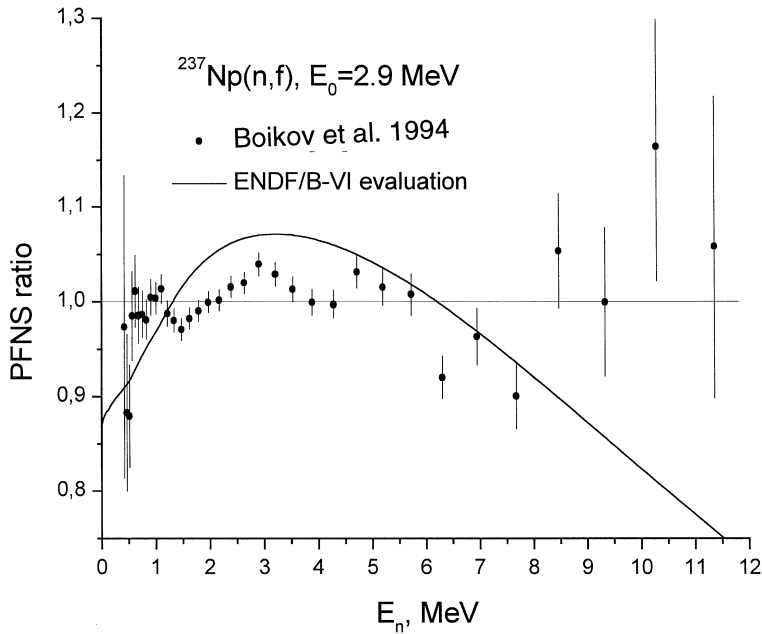


Fig. 11. The ratio of PFNS according to Boikov et al., 1994 and ENDF/B-6 to our evaluation. Residual chi-square per experimental point is $\chi^2/m = 2.1$ and 7.2 (ENDF/B-VI) for energy range 0.5–12 MeV, $m = 35$.

1. The PFN spectrum may be described as a sum of two Watt distributions for light and heavy fragments with equal contribution:

$$S(E, E_0) = 0.5 \sum_i W_i(E, T_i(E_0), \alpha), E_{vl} = \frac{A_h}{A_l A} \alpha TKE, E_{vh} = \frac{A_l}{A_h A} \alpha TKE \quad (6)$$

where T_i are parameters ($i = 1, 2$ denotes light and heavy fragments), E_0 is the incident neutron energy, α is the ratio of the kinetic energy at the moment of the neutron emission to full acceleration value (TKE). The two fragments model allows us to avoid the summing for many pairs of fragments and conserves the main correlation between T_i and E_{vi} .

2. It is a well known fact (see for example Gerasimenko and Rubchenya, 1986; Knitter et al., 1991) that the CMS neutron energy depends on fragment mass and $\langle \varepsilon_l \rangle \neq \langle \varepsilon_h \rangle$, hence one can not use $T_l = T_h$ in Eq. (6). Having in mind that we presume statistical equilibrium between nascent fragments one should conclude that T in Eqs. (5) and (6) is not the temperature of the fragment before neutron emission but the model parameter that connected with temperature. This may be explained by neutron cascade emission, because of statistical Hauser–Feshbach calculations assuming equilibrium between fragment (Gerasimenko and Rubchenya, 1986) reproduce experimental $\varepsilon(A_i)$ dependence (A_i – fragment mass number) reasonably well. For simplification

one may assume that T parameters in Eqs.(5) and (6) are only proportional to temperature t , $T_1 = c_1 \cdot t$, $T_h = c_h \cdot t$. It means that the ratio $T_1/T_h = \text{const}$ does not depend on incident energy and nucleus if c_1 , c_h are identical for any fissile system. These assumptions were checked for some nuclei and it was found $T_1/T_h = 1.248 \pm 0.031$.

3. The T parameters for heavy fragments were calculated relative to ^{252}Cf data according to the following formula:

$$T_h^x = T_h^{Cf} \sqrt{\frac{U^x A^{Cf}}{U^{Cf} A^x}}, \quad (7)$$

where $U = E_r + B_n + E_0 - TKE$, E_r - energy release, B_n - binding neutron energy, E_0 - incident neutron energy, A - mass number of fissile nucleus. The above relation was obtained under the following assumptions: excitation energy $U = a \cdot t^2$; $U = U_1 + U_h$; level density parameter $a = \text{const } A_i$; $A = A_1 + A_h$.

Recent investigations (Kornilov and Kagalenko, 1994, 1997) demonstrated that for ^{252}Cf spontaneous fission the fragment kinetic energy at the moment of neutron emission E_v is less than E_{v0} , calculated from TKE. We used this hint to incorporate free parameter α that connected with fragment kinetic energy in the Watt formula by the relation: $E_v = \alpha E_{v0}$ [see Eq. (6)]. One should emphasize that it is a semi-empirical model that contains free parameter fitted to the experimental PFNS. Therefore, α values depends on model assumptions: Eq. 5–7, T_1/T_h ratio, TKE and one should use the self-consistent set of parameters presented by Kornilov et al., 1999 for the practical application.

The mentioned above assumptions were checked on large set of the experimental data. It was shown that the proposed systematic provides a more accurate description of the experimental data than Madland–Nix model. The similar conclusion is demonstrated in this paper.

The fitted $\alpha = 0.799 \pm 0.015$ at 0.52 MeV incident neutron energy agrees very well with $\alpha = 0.817 \pm 0.007$ estimated from Boikov et al. (1994) data at 2.9 MeV. In both cases the experimental data and evaluated spectra show similar deviation from ENDF/B-VI data.

The PFNS were also measured for 1–2.5 MeV (seven points) incident neutron energies. However, the rather narrow energy range from E_0 to ~ 7 MeV was available for fission spectra analysis at this incident energy range. The low energy limit is determined by the contribution of the elastic and inelastic neutron scattering. At $E > 7$ MeV the statistical errors were more then 50%. It resulted in a large spread of the fitted α -values. The average fitted α -value for this energy interval was $\langle \alpha \rangle = 0.8 \pm 0.11$ (the error is the mean root spread of the fitted data) agrees reasonably well with data at 0.52 MeV. The evaluated spectrum shape also agrees inside error bars with the experimental data for incident energies 1–2.5 MeV.

The experimental $\langle E \rangle = 1.96 \pm 0.04$ was calculated by numeric integration of the experimental points and was corrected for finite energy interval (0.6–11 MeV)

using the evaluated spectrum shape. The experimental average neutron energies $\langle E \rangle$ of this work, Boikov et al. (1994) and Trufanov et al. (1992) together with the evaluated curve are shown in Fig. 12. The analysis of the experimental data from Trufanov et al. allows us to conclude that this work overestimates the contribution of high energy neutrons (> 6 MeV). The similar conclusion can be made from the comparison of experimental and theoretical results presented in Trufanov et al. (1992). It seems that the experimental data of this work has a systematic error that may explain the increase of the average energy in Fig. 12.

So, the new PFNS evaluation¹² for ^{237}Np based on our experimental data for 0.52 MeV and data of Boikov et al. (1994) was used in following analysis for estimation of the inelastic scattering spectra.

4. Inelastic neutron scattering on ^{237}Np

As was shown in previous section the “monoenergetic” neutron source has a low energy background due to the scattering on the target environment and tritium admixture into the target backing. The share of the background neutrons is rather small, however, being scattered by the sample nuclei (Np, O, Fe) they may produce the contribution comparable with inelastic scattering on Np. So, the detailed simulation of the experiment, bearing in mind all the above mentioned peculiarities, is the crucial point of this work. The Monte Carlo code BRAND (Androsenko et al., 1985) was applied for data evaluation. The MC calculation results were used to estimate the neutron flux on the sample, to correct the data for oxygen and iron admixture, the

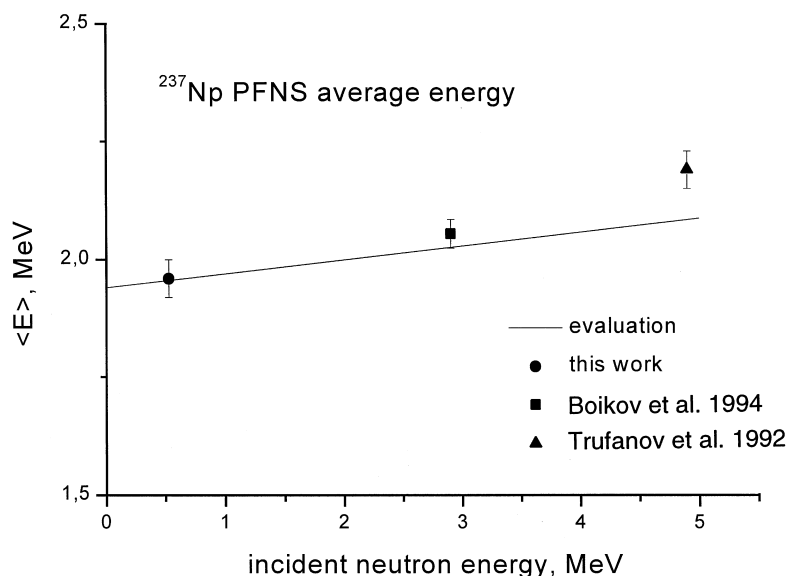


Fig. 12. The average energy for fission neutrons as a function of incident neutron energy.

contribution of the “tail” neutrons and the finite size of the sample. The MC results calculated in time scale were transformed to energy with the procedure applied for the experimental data. The neutron cross-sections for Np, Fe, C, O and H nuclei required for MC simulation were taken from ENDF/B-VI evaluation.

4.1. Data correction and normalization

The comparison of experimental data and calculated results for carbon (Fig. 13) allow us to conclude that the response function of the spectrometer is predicted with reasonable accuracy to subtract the neutron scattering on the oxygen and iron nuclei admixture in the Np-sample. In addition these data confirm our conclusion that an additional source of the neutrons is required. The model of the “monoenergetic” source only (the scattering on the target environment was taken into account) can not describe the experimental data.

The goal of the data correction may be defined as follow. One should transform the experimental data $N_{\text{exp}}(E)$ to the ideal spectrum $I(E)$ that may be described by equation:

$$I(E) = \Phi_0 < l > \rho \frac{d^2\sigma}{dE d\Omega} \quad (8)$$

where Φ_0 is total amount of “monoenergetic” neutrons hit by the sample, $< l >$ is the average length of the sample along neutron trajectory, ρ is a sample density, $d^2\sigma/dE d\Omega$ —investigated cross-section.

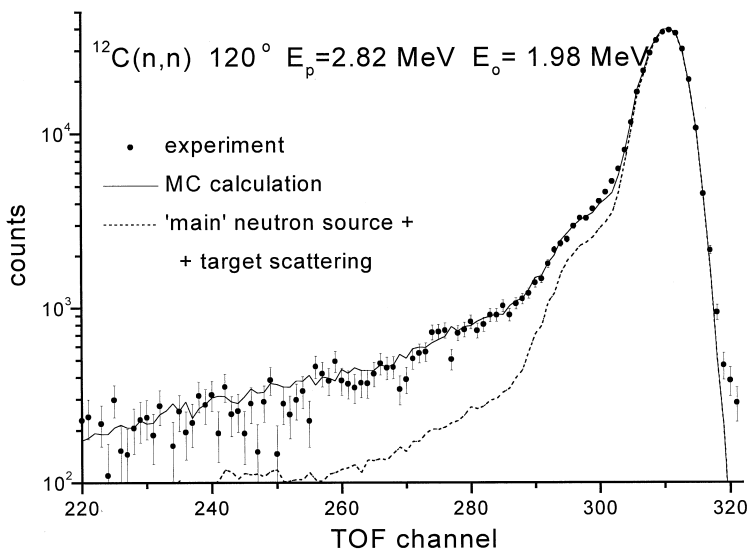


Fig. 13. The TOF spectrum of neutrons scattered by carbon sample for 1.98 MeV incident neutron energy. Lines show results of the Monte Carlo simulation.

Due to the fact that there exist two independent sources of neutrons the measured spectrum should be described by the equation:

$$N_{\text{exp}}(E) = \Phi_0 \left[(1 - \beta) F_0(E_0, E) + \beta \frac{\Phi_1}{\Phi_0} F_1(E_1, E) \right] \quad (9)$$

where 0,1 indexes denote Ti- layer and target backing, $F_i(E_i, E)$ are responses of the spectrometer for one neutron with energy E_i hitting the sample, including scattering on the target environment, multiple scattering and attenuation in the sample, scattering on the admixture nuclei, sample size and detector properties. These functions transform the neutron energy from E_i to E and consist of proper sums and integrals. The total amount of neutrons exposed in the sample may be presented by the relation $\Phi = (1 - \beta)\Phi_0 + \beta\Phi_1$, where β is the share of tritium into the target backing.

The calculated spectrum may be described by the similar equation:

$$N_{\text{calc}}(E) = (1 - \beta) F_0(E_0, E) + \beta \frac{\omega_1}{\omega_0} F_1(E_1, E) \quad (10)$$

where ω_i are the average weights of neutrons hit the sample which depends on the T(p,n) reaction cross-section and amount of tritium in the target. They are proportional to Φ_i from Eq. (9). The structure of Eq. (10) has the following explanation. In the case of two sources which have different yield and energy distribution one should normalize the MC result to one proton exposed to the target. Therefore, the denominator in Eq. (10) gives the amount of neutrons and re-normalizes the data to one “monoenergetic” neutron. Hence, one should make two independent calculations for tritium into Ti-layer and target backing normalized to the same amount of tritium summing the result with coefficient β to construct the spectrum according to Eq. (10).

For elastic scattering on carbon $F_i(E_i, E)$ functions may be calculated with high accuracy. The β value may be adjusted from the comparison experimental and calculated spectra (see Fig. 13). So, the ratio of experimental spectrum to calculated one for C- sample gives Φ_0 which should be put in Eq. (8) to calculate absolute cross-section for Np.

The differential cross-section for Np was estimated from the following equation:

$$\frac{d^2\sigma}{dE dX} = k_{\text{norm}} \cdot \frac{N_{\text{calc}}^0}{N_{\text{calc}}} \cdot N_{\text{exp}} \quad (11)$$

where N_{exp} is the experimental spectrum corrected for neutron detector efficiency, $k_{\text{norm}} = (\Phi_0 < l > \rho)^{-1}$ is the normalization coefficient, ρ is a ^{237}Np density in the sample, N_{calc} is the calculated spectrum that simulates real experiment in all details according to Eq. (10), and N_{calc}^0 is the spectrum that simulates the hypothetical situation- “Np only” sample with real Np density, but without multiply scattering and attenuation and without the “tritium backing” neutrons [Eq. (8)]. The average sample length $< l >$ and ω_0 [see Eq. (10)] was calculated simultaneously with N_{calc} .

The β values were fitted to the carbon spectra, that were measured just after the Np-runs. So, we should assume that the β value does not change from Np- to C- run.

4.2. Errors analysis

The following factors were taken into account to estimate total errors of the experimental data:

1. Statistical uncertainties — the statistical uncertainties changed from ~ 1 to $> 50\%$ depending on neutron energy. The highest error was for fission neutrons in energy range > 7 MeV.
2. The uncertainties of the data normalization and sample position — the samples fixed in the sample holder with error ± 1 mm. It gives $\sim 2\%$ error at the distance 10 cm from the source to the sample. The normalization procedure was checked by comparison of the experimental number of fission neutrons $(\sigma_f \cdot v)$ with evaluated values. The experimental values were estimated by fitting the evaluated fission neutron spectra with experimental data in the high energy region of spectrum $E > E_0$. The ratios r of the experimental $(\sigma_f \cdot v)_{\text{EXP}}$ and evaluated $(\sigma_f \cdot v)_{\text{ENDF/B-VI}}$ values are presented in Table 1. The average $\langle r \rangle = 1.016 \pm 0.033$ (the error δr is a mean root spread of the data). The $\delta r = 0.033$ estimated from this comparison is a real experimental uncertainty of the normalization procedure including the sample position uncertainty. It is rather difficult to estimate the uncertainties of the evaluated $v\sigma$ value. It seems that they are known to an accuracy of $\sim 2\%$. So, the average $\langle r \rangle = 1.016$ does not contradict to this value. Two additional factors were included in the total error: $\pm 1\%$ due to carbon elastic scattering cross-section and $\pm 2\%$ due to the error for the number of the Np-nucleus in the sample. It should be emphasized that the omission of the “tritium backing” neutrons changes the normalization to $\sim 5\%$.
3. Detector efficiency uncertainties — the relative efficiency is required for our data processing. It was calculated relative to ^{252}Cf spectrum which is known in an accuracy of 1.5–1.8% ($E = 0.2$ –1 MeV), $\sim 1.2\%$ (1–4 MeV) and $\sim 2\%$ (4–10 MeV). Having in mind the experimental procedure, the error of the detector

Table 1

The experimental $(\sigma_f v)$ values and their ratio to ENDF/B-VI data. The errors of the fitting procedure are given in brackets. The systematic errors due to normalization were not included

E_0 , MeV	$(\sigma_f v)_{\text{exp}}$	$(\sigma_f v)_{\text{exp}}/(\sigma_f v)_{\text{eval}}$
0.99	3.89 (1.3%)	0.963
1.00	4.00 (1.3%)	0.989
1.46	4.53 (1.1%)	1.010
1.49	4.50 (1.2%)	0.999
1.77	5.02 (1.0%)	1.055
1.99	5.05 (1.0%)	1.031
2.47	5.25 (1.2%)	1.058

efficiency was estimated as $\sim(20\text{--}10)\%$ ($E = 0.2\text{--}0.6$ MeV), $\sim 2\%$ ($0.8\text{--}4$ MeV) and $> 5\%$ ($4\text{--}10$ MeV).

The statistical and systematic uncertainties mentioned earlier were combined in quadrature.

4.3. Inelastic neutron spectra and cross sections

The composed effect due to interaction of the incident neutrons with the ^{237}Np sample was measured in our experiment. Therefore, the evaluated inelastic spectra and the inelastic cross-section depends on the neutron data library used for calculation of the correction. The experimental data $k_{\text{norm}} \cdot N_{\text{exp}}$ and calculated spectra N_{calc}^o , N_{calc} are shown in Figs. 14–16 for an energy range of the Np elastic and inelastic scattering $E < E_0$. One can see a rather smooth distribution and three peaks due to neutron scattering on Np, O and Fe. One may conclude that a reasonable agreement exists between experimental data and the N_{calc} spectra. It may prove the correction procedure determined by Eq. (11). Secondary, we can predict the total energy resolution of the spectrometer (see also Figs. 8 and 13) with high accuracy. It gave us a basis for the correct subtraction of the “peak-like” background due to scattering on oxygen and iron.

The following peculiarities of the experimental data should be highlighted. The elastic scattering on oxygen is of particular importance for the calculation of the correction at ~ 1 MeV initial neutron energy. According to Fig. 16 the ENDF/B-VI data underestimate the elastic scattering (peak at $0.7\text{--}0.8$ MeV) at a 120° angle. The experimental data should be reduced by 0.86 (or one should increase the calculated

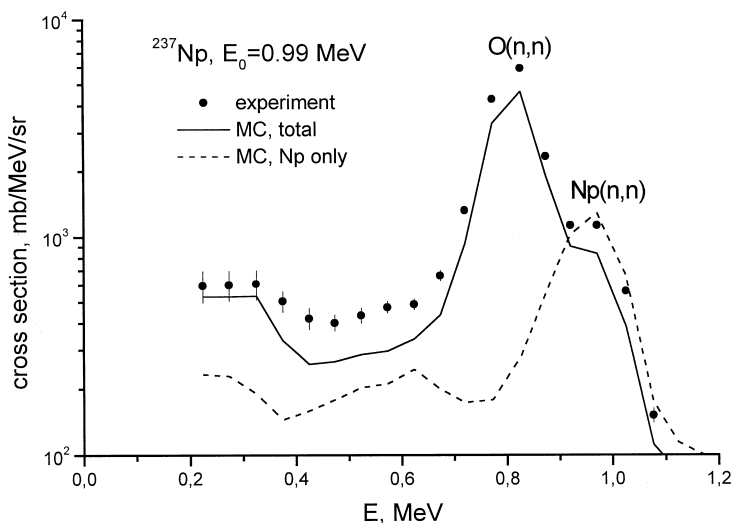


Fig. 14. Experimental data as $k_{\text{norm}}N_{\text{exp}}$ according to Eq. (11) and calculated spectra N_{calc}^o (solid line) and E_{calc}^o , (dashed line) for $E_0 = 0.99$ MeV.

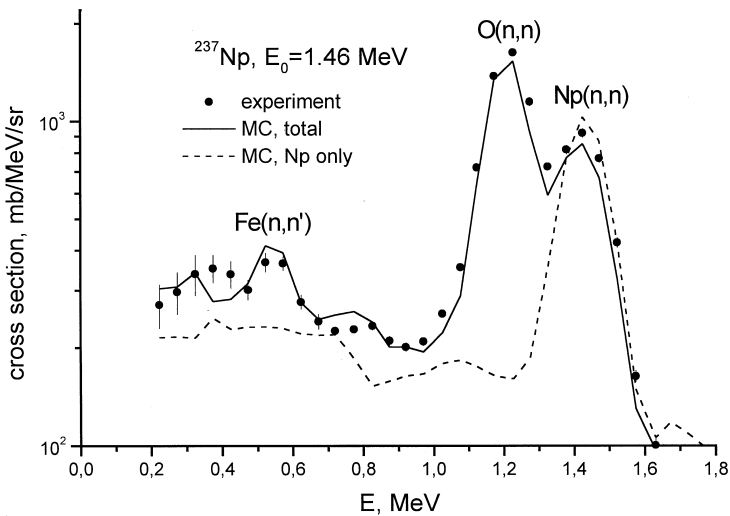


Fig. 15. The same as in Fig. 14 for $E_0 = 1.46$ MeV.

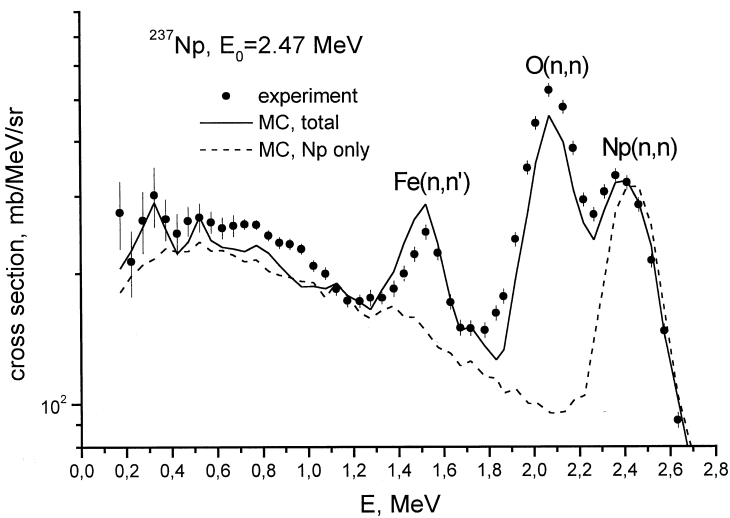


Fig. 16. The same as in Fig. 14 for $E_0 = 2.47$ MeV.

data). However, we cannot explain the difference due to data normalization (see Table 2). Hence it may be connected with the wrong neutron data. So, the rather large difference between experimental and calculated results in the energy range < 0.7 MeV in Fig. 14 may be connected with the underestimation of the “tail” neutron scattering. The elastic scattering by oxygen, to backward angle decreases with increasing of the initial neutron energy (see Figs. 14–16). In addition, there are a good agreement between experimental results and calculated data for 1.5–2 MeV

Table 2

The inelastic cross-section integrated in the energy range from 0.25 MeV to $(E_0 - 0.25)$ MeV

E_0 , MeV	$d\sigma/dX$, mb/sr
0.99	63.8 ± 9.1
1.00	70.4 ± 9.2
1.46	57.6 ± 6.0
1.49	59.1 ± 6.4
1.77	73.8 ± 7.4
1.99	83.8 ± 6.8
2.47	90.5 ± 6.1

energy range. So, the oxygen data uncertainty may distort the result at ~ 1 MeV energy only.

It seems, that the evaluated data from the ENDF/B-VI file overestimate the inelastic scattering on the first level for ^{56}Fe (the difference in the peak region at ~ 1.5 MeV in Fig. 16). The ratios of the experimental to calculated peak intensities are ~ 0.7 , 0.9 and 0.6 for incident energies ~ 1.5 , 2 and 2.5 MeV. This may reduce inelastic cross-section near iron peak region. However, the contribution of these energy intervals to the energy integrated cross-section is rather small and does not change the final result very much.

The differential cross-section for secondary neutron emission from Np $\frac{d^2\sigma}{dEdX} = \text{inelasc} + \text{elastic} + \text{fission}$ was calculated according to Eq. (11). The PFN spectra calculated as it was discussed in Section 3.3 were normalized to experimental data in the energy range $E > E_0$ and were subtracted from experimental emission data in the energy range $E < E_0$. In such a way, $d^2\sigma/dEdX$ differential elastic and inelastic cross-section at 120° was estimated.

Our data for inelastic and elastic scattering are presented in Figs. 17–19 together with ENDF/B-VI, JENDL-3 and BROND-3 data (inelastic spectra only). The results measured in independent runs for Mo and Cu target backing at incident energies $E_0 \sim 1$ MeV and ~ 1.5 MeV agrees inside the experimental error (see Figs. 17 and 18). It confirms that there are no large systematic errors due to the data normalization and the estimation of the neutron “tail” contribution. It also confirms the stability of the spectrometer during long time experiment. Our data confirms the existing of the peaks in the inelastic spectra of ^{237}Np connected with group of levels at 0.2–0.5 MeV excitation energy.

The inelastic neutron spectrum may be measured only in the limited energy interval. Some amount of neutrons is lost due to detector threshold (< 0.2 MeV) another one due to the scattering on the first level of Np in the energy range $\sim E_0$ that can not be separated from elastic neutrons. This fact complicate the direct comparison of the experimental and evaluated inelastic cross-section. To avoid this problem the spectra were integrated in the energy range from 0.25 MeV to $(E_0 - 0.25)$ MeV. The differential cross sections at 120° after such an integration are shown in Fig. 20 and presented in Table 2. Our data demonstrate the regular increase of the inelastic cross-section from 1.5 to 2.5 MeV that is not reproduced by the evaluated data. In

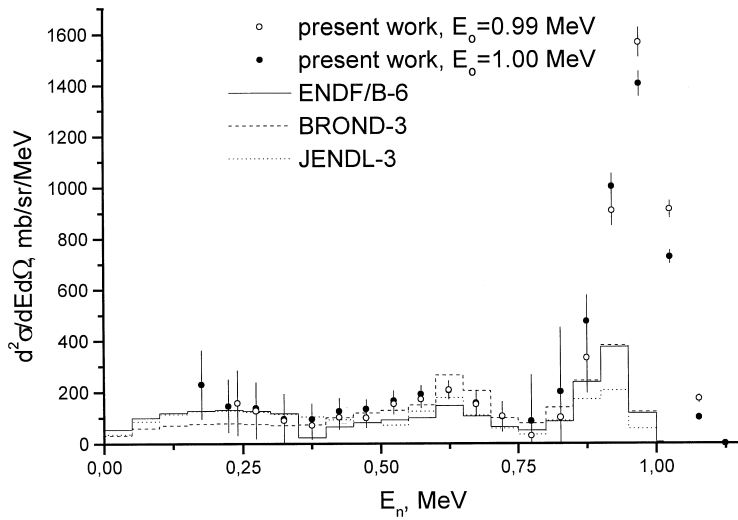


Fig. 17. Experimental spectra of inelastic and elastic scattering neutrons (points) for $E_0 \sim 1$ MeV incident neutron energy. Histograms show the inelastic spectra only according to various data libraries.

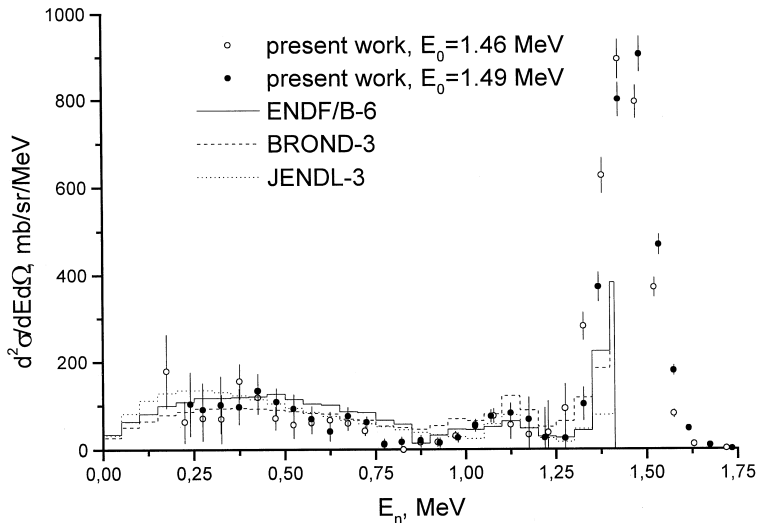


Fig. 18. The same as in Fig. 17 for $E_0 \sim 1.5$ MeV.

addition, one may speak about the existence of the broad minimum for the inelastic cross-section at energy range from 1 to 2 MeV. However, having in mind the difficulties connected with data measurement and data analysis one may conclude the reliable agreement between the BROND-3, the JENDL-3 and the experimental results. The curves for evaluated data in Fig. 17–20 were prepared with program packages GRUCON and REBUS (Sinititsa and Rineyskiy, 1993).

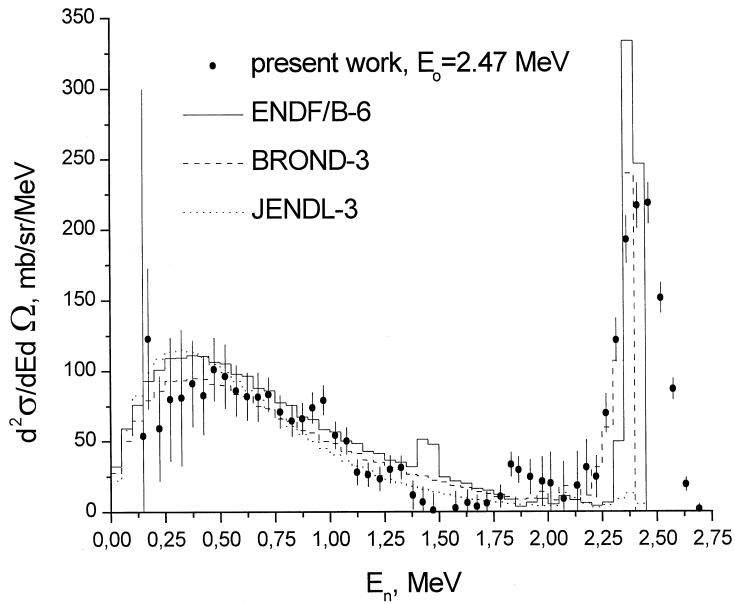


Fig. 19. The same as in Fig. 17 for $E_0 \sim 2.47$ MeV.

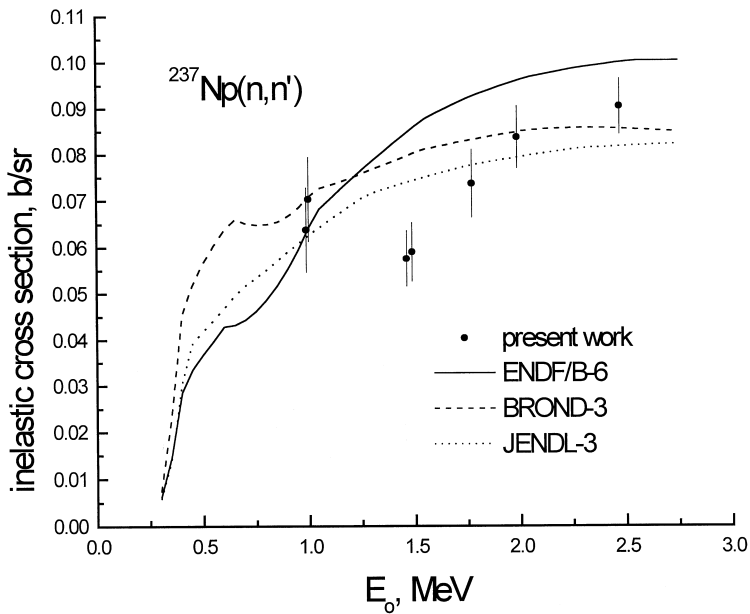


Fig. 20. Inelastic cross-section integrated over the energy range from 0.25 MeV to $(E_0 - 0.25)$ MeV for 120° emission angle versus incident neutron energy.

5. Conclusion

The analysis of the experimental data measured in this work allows us to conclude that the ENDF/B-VI data can not describe the ^{237}Np PFNS with required accuracy. Therefore, the result of new evaluation was applied for subtraction of the PFNS contribution.

The measurement of the inelastic neutron spectra with oxide sample is a very difficult task. The successful realization became possible due to the detail investigation of the neutron source properties and careful simulation of the experiment by Monte Carlo method. The omission of the “tail” neutrons will increase to ~ 2 times the measured inelastic cross-section.

The experimental data for inelastic neutron scattering supports the BROND-3 and the JENDL-3 data libraries. The comparison with other experimental results is impossible because of the inelastic cross-section for ^{237}Np have not been measured before.

Acknowledgements

We would like to thank V.V Sinitsa for data calculation and EG-1 staff for good work of the accelerator.

References

- Androsenko, A.A., Androsenko, P.A. et al., 1985. Code complex BRAND for radiation transport calculation by the Monte Carlo method. VANT (Voprosi atomnoi Nauki i Tekhniki), ser Fizika i Tekhn. Yadernykh Reaktorov 7, 33.
- Baryba, V., Ya, Kornilov, N.V., Semenova, N.N., 1977. ^{252}Cf neutron source with timing of neutron production. VANT (Voprosi atomnoi Nauki i Tekhniki) 5 (19), 45, (in Russian).
- Birukov, N.S., Kanaki, V.N., Triкова, V.I., 1990. Pulse mode of the EG-1 accelerator. Preprint FEI-2132, Obninsk (in Russian).
- Boikov, G.S. & Dmitriev, V.D. et al., 1994. Neutron spectra in ^{237}Np fission induced by neutrons of energy 2.9 and 14.7 MeV. Physics of Atomic Nuclei 57(12), 2047 (translated from Yadernaya Fizika).
- Gerasimenko, B.F., Rubchenya, V.A., 1986. Theoretical calculation of prompt fission neutron spectrum for ^{252}Cf spontaneous fission. In: Okamoto, K. (Ed.), Proceedings of an Advisory Group Meeting on Properties of Neutron Sources, Leningrad, June 1986, Report IAEA-TECDOC-410, Vienna, p. 208.
- Knitter, H.-H., Brosa, U., Budtz-Jorgensen, C., 1991. Neutron and gamma emission in fission. In: Wagemans, C. (Ed.), The Nuclear Fission Process. CRC Press, p. 497.
- Kornilov, N.V., 1986. Possibility of experimental determination and theoretical prediction of different properties of accelerator based neutron sources. In: Okamoto, K. (Ed.), Proceedings of an Advisory Group Meeting on Properties of Neutron Sources, Leningrad, June 1986, Report IAEA-TECDOC-410, Vienna, p. 230.
- Kornilov, N.V., Kagalenko, A.B., Baryba, V.Ya. et al., 1991. Time of flight neutron spectrometer on the EG-1 accelerator. Preprint FEI-2174, Obninsk (in Russian).
- Kornilov, N.V., Kagalenko, A.B. 1994. Pre-acceleration neutron emission and prompt fission neutron spectra. Journal of Nuclear Physics, 57(7), 1240; Pre-acceleration neutron emission — is it evaluation trick or reality. Proceedings of ISINN-2, Dubna, p. 116.

- Kornilov, N.V., Kagalenko, A.B., 1995. Inelastic neutron scattering by ^{235}U and ^{238}U nuclei. *Nuclear Science and Engineering* 120, 55.
- Kornilov, N.V., Kagalenko A.B., 1996. Inelastic neutron scattering on the first level of ^{238}U . *Proceedings of ISINN-4*, Dubna, p. 245.
- Kornilov, N.V., Kagalenko, A.B., 1997. On the mechanism of neutron emission in fission. *Proceedings of Trieste Conference on Nuclear Data for Science and Technology* 1, 179.
- Kornilov, N.V., Kagalenko, A.B., Baryba V.Ya., et al., 1997. Measurement and analysis of the basis nuclear data for minor actinides, In: Rabotnov, N.S. (Ed.), *Final report ISTC-304*, Chapter III.
- Kornilov, N.V., Kagalenko, A.B. et al., 1998. The tritium admixture in solid target and neutron spectrum from 'monoenergetic source'. Preprint IPPE-2720, Obninsk (in English).
- Kornilov, N.V., Kagalenko, A.B., Hamsch, F.-J., 1999. Calculation of the prompt fission neutron spectra on the basis of new systematic of the experimental data. *Journal of Nuclear Physics* 62 (2), 173.
- Le Couter, K.J., Lang, D.W., 1959. Neutron evaporation and level densities in excited nuclei. *Nuclear Physics* 13, 32.
- Madland, D.C., Nix, J.R., 1982. New calculation of prompt fission neutron spectra and average prompt neutron multiplicities. *Nuclear Science and Engineering* 81, 213.
- Mannhart, W. 1986. Evaluation of the Cf-252 fission neutron spectrum between 0 and 20 MeV. In: Okamoto, K. (Ed.), *Proceedings of an Advisory Group Meeting on Properties of Neutron Sources*, Leningrad, June 1986, Report IAEA-TECDOC-410, Vienna, p. 158.
- Sinitsa, V.V., Rineyskiy, A.A., 1993. GRUCON — a package of applied computer programs. INDC(CECP)-344, IAEA, Vienna.
- Trufanov, A.M. et al., 1992. Energy distribution of ^{237}Np fission neutrons. *Journal of Nuclear Physics* 55 (2), 289.